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COURSE CODE: ECA0910
COURSE NAME: DIGITAL SIGNAL PROCESSING
PRACTICAL LAB RESULTS

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EXP: 2.1:
CODE:
clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);

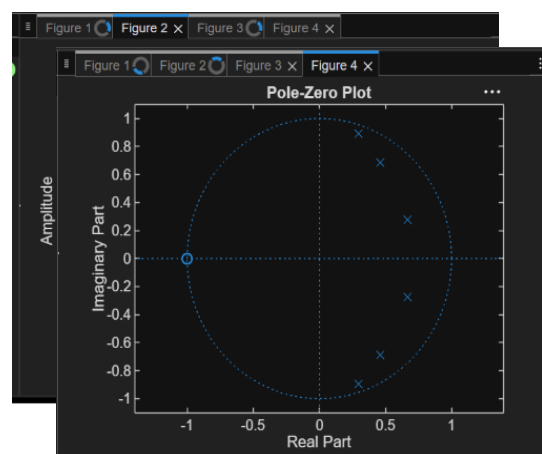
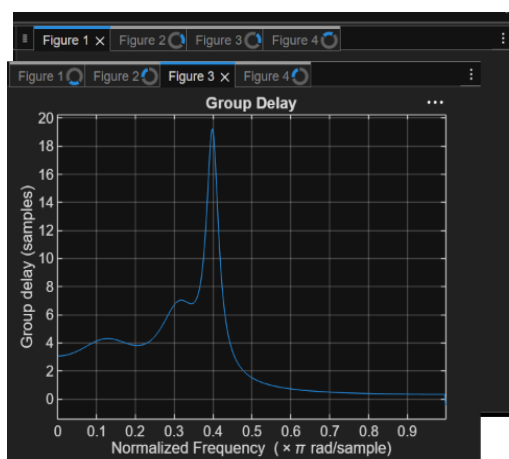
[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

% ----- GRAPHS -----

figure; freqz(b,a); % Magnitude + Phase
figure; impz(b,a); % Impulse Response
figure; grpdelay(b,a); % Group Delay
figure; zplane(b,a); % Pole-Zero Plot
OUTPUT:
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EXP:2.2:
CODE:
clc;

clear;

close all;

% Specifications
Rp = 1; % Passband Ripple (dB)
Rs = 40; % Stopband Attenuation (dB)
Wp = 0.5; % Passband Edge
Ws = 0.35; % Stopband Edge

% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');

% Impulse Response
figure;
impz(b,a);
title('Impulse Response');

% Group Delay
figure;
grpdelay( b,a);
title('Group Delay');

% Pole-Zero Plot
figure;

```

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zplane(b,a);

title('Pole-Zero Plot');

fprintf('Filter Order = %d\n',n);

fprintf('Cutoff Frequency = %.4f\n',Wn);

```

OUTPUT:

